

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**

Off-Campus Centre of Nitte (Deemed to be University)

First Semester B.Tech (CBCS) Degree Examinations

December 2023

CY1005-1 – CHEMISTRY OF ENERGY STORAGE AND DISPLAY DEVICES

(For ACT, ECE, EEE, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The drawback of primary cell is that
A) Chemical action is not reversible
B) Electrolyte used is very costly
C) Chemical action is reversible
D) It is not portable
2. Chemical energy is converted to energy by a fuel cell.
A) Solar
B) Potential
C) Electrical
D) Mechanical
3. Most widely used solar cell material is
A) Mild Steel
B) Silicon
C) Mercury
D) Iron
4. The majority of charge carriers in P-type semiconductor are.....
A) Electrons
B) Neutrons
C) Holes
D) None of the above
5. The process of heat softening, moulding and cooling to rigidity can be repeated for which plastics?
A) Thermoplastics
B) Thermosetting
C) both (a) and (b)
D) neither (a) nor (b)
6. A high molecular weight molecule that is built from a large number of small molecules is known as a.....
A) Polymer
B) Isomer
C) Monomer
D) Tautomer
7. is the main principle of electroless plating.
A) Hydrolysis
B) Autocatalytic reduction
C) Electrolysis
D) Esterification
8. Potential of the calomel electrode depends on concentration of.....
A) Hg
B) KCl
C) Hg_2Cl_2
D) KOH
9. pH of the solution can be determined using the electrode combination
A) Glass-calomel electrode
B) Nickel-Calomel electrode
C) Glass-Zinc electrode
D) Zinc- Calomel electrode
10. Corrosion between the dissimilar metals is called as.....
A) Concentration cell corrosion
B) Oxidation corrosion
C) Dry corrosion
D) Galvanic corrosion
11. Corrosion due to the formation of cavities around the metal is called as the.....
A) Water line corrosion
B) Pitting corrosion
C) Soil corrosion
D) Galvanic corrosion

12. The function of a sensor is to.....
 A) Detect events within specified environment B) Separate physical parameters
 C) Track and transfer data to computer processor D) Both a and c
13. A temperature sensor is provided with as input.
 A) Sound B) Liquid
 C) Air D) Temperature
14. are the electrodes used for potentiometric estimation of FAS.
 A) Platinum-calomel electrode B) Platinum-glass electrode
 C) Zinc-calomel electrode D) Calomel-calomel electrode
15. Materials having two dimensions in nano scale range are called..... nanomaterials
 A) Zero Dimensional B) One Dimensional
 C) Two Dimensional D) Three Dimensional
16. CVD stands for.....
 A) Carbon Vapor Dissolution B) Carbon Vapor Deposition
 C) Chemical Vapor Deposition D) Chemical Vapor Dissolution
17. Liquid crystal molecules possess.....
 A) Vibrational order B) Intrinsic order
 C) Transational order D) Orientational order
18. Which of the following televisions delivers the best picture quality?
 A) LCD B) OLED
 C) Plasma D) 3D
19. Which of the following is one of the best solutions to get rid of nonbiodegradable waste?
 A) Recycling B) Burning waste
 C) Burying waste D) All the above
20. E-waste causes..... pollution.
 A) Soil B) Air
 C) No D) Sound

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the construction and working of Li- ion battery during discharging.	6	L2	1	1
	b) Describe the construction and working principle of proton exchange polymer membrane fuel cell.	6	L2	1	1
	c) Give any four advantage of solar PV cell.	4	L1	1	1
2.	a) Calculate the number average and weight average molecular mass of a polymer sample in which 30% molecules have a molecular mass 20,000, 40% have 30,000 and rest have 60,000.	6	L3	1	2
	b) Explain the conducting mechanism of polyacetylene with reactions.	6	L2	1	1
	c) Define the following: i) graft copolymer, ii) degree of polymerization, iii) polymer composite, iv) elastomers.	4	L1	1	1
3.	a) Describe the construction and working of solar PV cell.	6	L2	1	1
	b) Explain the synthesis and application of following polymers. i) Silicon rubber, ii) epoxy resin.	7	L2	1	1
	c) Give any three advantages of electroless plating.	3	L1	1	1
Unit – II					
4.	a) Explain the construction and working of Calomel electrode.	6	L2	2	1
	b) The spontaneous galvanic cell, Sn/SnSO ₄ (0.024M)// SnSO ₄ (0.064M)/Sn, develops an emf of 0.0126 V at 25°C. Calculate the valency of Tin.	4	L3	2	2
	c) Explain the working principle and application of optical sensor.	6	L2	2	1

5. a)	Explain the electrochemical theory of corrosion by taking iron as an example.	7	L2	2	1
b)	Discuss the principle & experimental method of conductometric estimation of weak acid	7	L2	2	1
c)	Define the following: i) differential metal corrosion, ii) galvanization.	2	L1	2	1
6. a)	Explain the construction of glass electrode. How to determine the pH of unknown solution using a glass electrode.	7	L2	2	1
b)	Explain the steps involved in the galvanization process.	3	L2	2	1
c)	Explain the working principle and application of electrochemical sensor.	6	L2	2	1

Unit – III

7. a)	Describe the synthesis of nanomaterials by sol-gel method.	5	L2	3	1
b)	Explain the properties and application of Perovskite materials.	6	L2	3	1
c)	Explain the extraction of gold from e-waste.	5	L2	3	1
8. a)	Describe the synthesis and applications of nanotubes by CVD method.	6	L2	3	1
b)	Give any four properties of quantum light emitting diodes.	5	L1	3	1
c)	What is e-waste? Give any four advantages of recycling.	5	L1	3	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
